

Reproduce-then-Extend — State Policy Liberalism (American State Politics) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Caughey, D. & Warshaw, C. (2016). “The Dynamics of State Policy Liberalism, 1936–2014.” *American Journal of Political Science* 60(4):899–913. Building on the state-representation tradition (Erikson, Wright & McIver, *Statehouse Democracy*, 1993), the paper measures each state’s **policy liberalism** and links it to public opinion and state political conditions – a **cross-sectional OLS** of policy liberalism on state characteristics. This example illustrates **regularization and variable selection** in a small-sample ($n = 50$) design, where predictors are many relative to observations.

1 Background

A central question in American state politics is why some states adopt systematically more **liberal policies** than others. The state-representation literature argues that policy tracks **public opinion** (the “opinion–policy link”), mediated by **party control** and the organized strength of liberal and conservative constituencies. Caughey & Warshaw construct a dynamic measure of state policy liberalism and show it responds to mass preferences. Here we treat the **cross-section of the 50 states** as the unit of analysis and ask which state characteristics predict policy liberalism – the setting the state-politics literature has studied with ordinary regression for decades.

2 Data and codebook

Unit of analysis: a U.S. **state** (cross-section averaged over 2005–2012); $n = 50$. Data are drawn from the **Correlates of State Policy Project** (Jordan & Grossmann 2020), which compiles the standard state-politics measures.

Variable	Definition
pollib_median	state policy liberalism (Caughey–Warshaw), higher = more liberal (outcome)
mood	public policy mood (higher = more liberal preferences)
citi6013	citizen ideology (Berry et al.), higher = more liberal
ranney4_control	Ranney index of Democratic party control (higher = more Democratic)
union_density	percent of workers unionized
evangelical_pop	evangelical share of the population
gini_coef	income inequality (Gini coefficient)
gsppcap	gross state product per capita

3 Descriptive results

```
d <- read.csv("data/state_policy_liberalism.csv")
preds <- setdiff(names(d), c("st", "pollib_median"))
cat(sprintf("%d states x %d predictors (p/n = %.2f)\n", nrow(d), length(preds),
  ↪ length(preds)/nrow(d)))

## 50 states x 7 predictors (p/n = 0.14)

round(t(sapply(d[c("pollib_median", preds)],
  function(x) c(mean = mean(x), sd = sd(x), min = min(x), max = max(x)))),
  ↪ 2)

##               mean      sd      min      max
## pollib_median   0.06   1.29   -2.27    2.44
## citi6013       53.36  14.70   23.34   86.52
## ranney4_control  0.49   0.31    0.00    1.00
## evangelical_pop 19.55  13.24    2.90   71.40
## union_density  11.23   5.60    3.14   24.91
## gini_coef       0.61   0.03    0.56    0.68
## mood           41.23   5.96   31.14   63.17
## gsppcap        41596.20 7747.83 28955.83 63450.17

par(mfrow = c(1, 2)); par(mar = c(4, 4, 2.5, 1))
hist(d$pollib_median, breaks = 12, col = "#a6cee3", border = "white",
  main = "Outcome: policy liberalism", xlab = "policy liberalism")
cors <- sort(sapply(d[preds], function(x) cor(x, d$pollib_median)))
par(mar = c(4, 7, 2.5, 1))
barplot(cors, horiz = TRUE, las = 1, cex.names = 0.85,
  col = ifelse(cors > 0, "#1f78b4", "#e31a1c"), main = "Correlation with
  ↪ liberalism",
  xlab = "correlation"); abline(v = 0, col = "grey50")
```

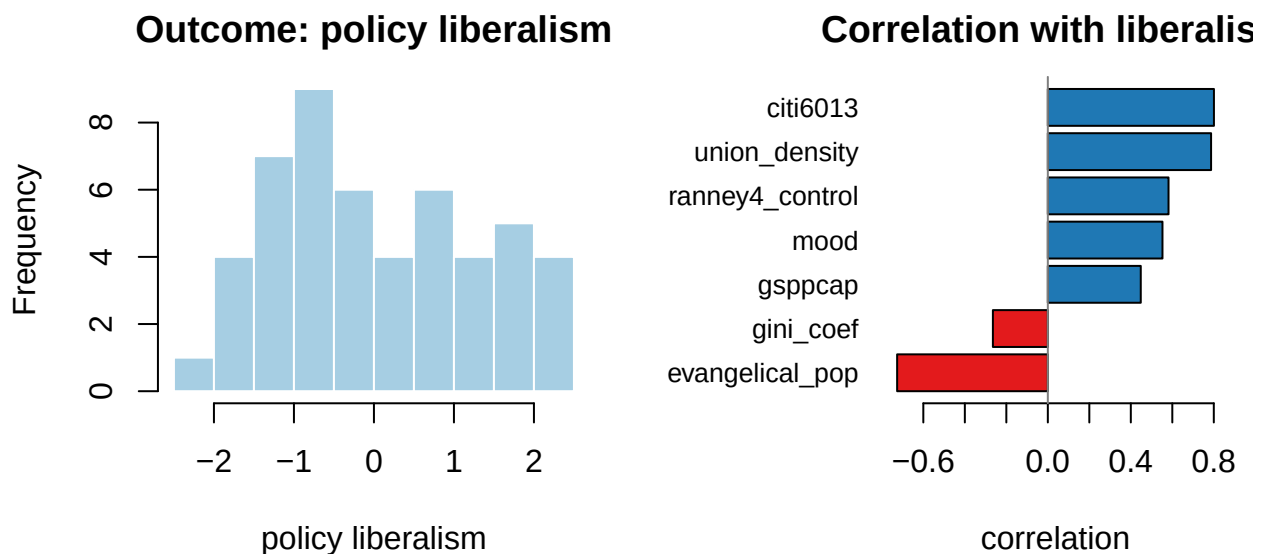


Figure 1: Policy-liberalism distribution (left) and each predictor's correlation with policy liberalism (right).

4 Reproduce the published regression

The state-representation model regresses policy liberalism on public opinion and state political conditions.

```
fit <- lm(pollib_median ~ ., data = d[, c("pollib_median", preds)])
round(summary(fit)$coefficients, 3)
```

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-3.195	2.180	-1.466	0.150
## citi6013	0.024	0.011	2.157	0.037
## ranney4_control	0.670	0.338	1.980	0.054
## evangelical_pop	-0.018	0.011	-1.642	0.108
## union_density	0.090	0.021	4.235	0.000
## gini_coef	0.001	3.072	0.000	1.000
## mood	0.019	0.019	0.991	0.327
## gsppcap	0.000	0.000	0.357	0.723

Confirmation against the literature. In-sample $R^2 = 0.81$, and the **opinion–policy link** is recovered: more liberal **public mood** and **citizen ideology**, stronger **Democratic control**, and higher **union density** all predict more liberal policy, while a larger **evangelical** population predicts more conservative policy – the signs the state-politics literature reports.

5 Regularization & variable selection

With only **50 states** and **7 predictors**, ordinary least squares has few observations per coefficient and **over-fits**: it fits the 50 states well but predicts new states poorly. Penalized regression – lasso (L_1), ridge (L_2), elastic net – shrinks the coefficients to trade a little bias for much lower variance.

```
X <- as.matrix(d[, preds]); y <- d$pollib_median
set.seed(2026)
lasso <- cv.glmnet(X, y, alpha = 1); ridge <- cv.glmnet(X, y, alpha = 0); enet <-
  cv.glmnet(X, y, alpha = 0.5)
n_keep <- sum(coef(lasso, s = "lambda.1se")[-1] != 0)
cat(sprintf("At lambda.1se the lasso keeps %d of %d predictors (the rest are set to
  exactly 0).\n",
            n_keep, length(preds)))
```

At lambda.1se the lasso keeps 4 of 7 predictors (the rest are set to exactly 0).

```
data.frame(
  OLS      = round(coef(fit)[-1], 3),
  Lasso     = round(as.numeric(coef(lasso, s = "lambda.1se"))[-1], 3),
  Ridge     = round(as.numeric(coef(ridge, s = "lambda.1se"))[-1], 3),
  ElasticNet = round(as.numeric(coef(enet, s = "lambda.1se"))[-1], 3),
  row.names = preds)
```

	OLS	Lasso	Ridge	ElasticNet
## citi6013	0.024	0.025	0.016	0.020
## ranney4_control	0.670	0.046	0.525	0.283
## evangelical_pop	-0.018	-0.007	-0.015	-0.013
## union_density	0.090	0.068	0.047	0.063
## gini_coef	0.001	0.000	-1.461	0.000
## mood	0.019	0.000	0.022	0.005
## gsppcap	0.000	0.000	0.000	0.000

Which fits best out of sample? With $n = 50$ a single split is noisy, so we **repeat a 70/30 split 50 times**: each replicate tunes the penalty by cross-validation on the training states and scores once on the

held-out states. Penalized models are scored at `lambda.min` (the loss-minimizing penalty).

```
rmse <- function(p, yt) sqrt(mean((yt - p)^2))
REPS <- 50; alphas <- c(Lasso = 1, Ridge = 0, ElasticNet = 0.5)
M <- matrix(NA, REPS, 4, dimnames = list(NULL, c("OLS", names(alphas))))
Xdf <- as.data.frame(X)
for (r in 1:REPS) {
  set.seed(2025 + r)
  te <- sample(nrow(X), round(0.30 * nrow(X))); tr <- setdiff(seq_len(nrow(X)), te)
  ols <- lm(y[tr] ~ ., data = Xdf[tr, ]); M[r, "OLS"] <- rmse(predict(ols, Xdf[te, ]),
  ↪ y[te])
  for (a in names(alphas)) {
    cv <- cv.glmnet(X[tr, ], y[tr], alpha = alphas[a])
    M[r, a] <- rmse(as.numeric(predict(cv, X[te, ], s = "lambda.min")), y[te])
  }
}
round(colMeans(M), 3) # mean held-out RMSE over 50
↪ splits
```

```
##      OLS      Lasso      Ridge ElasticNet
##      0.736      0.721      0.688      0.704
```

```
best <- names(which.min(colMeans(M)[names(alphas)]))
cat(sprintf("\nBest penalization: %s. Held-out RMSE: %s = %.3f vs OLS %.3f (%.0f%%
  ↪ lower).\n",
            best, best, min(colMeans(M)[names(alphas)]), colMeans(M)["OLS"],
            100 * (colMeans(M)["OLS"] - min(colMeans(M)[names(alphas)])) /
            ↪ colMeans(M)["OLS"])))
```

```
##
## Best penalization: Ridge. Held-out RMSE: Ridge = 0.688 vs OLS 0.736 (6% lower).
```

Because there are so few states per predictor, unpenalized OLS over-fits and the penalized models **predict new states more accurately**, while the lasso simultaneously **selects** the handful of state characteristics that carry the signal.

6 Best-subset selection with ABESS

Lasso reaches a sparse model through shrinkage. **Best-subset selection** instead searches directly for the subset of predictors that fits best; the **adaptive best-subset (ABESS)** algorithm does so efficiently and chooses the subset size automatically.

```
library(abess)
ab <- abess(X, y)
cat(sprintf("ABESS selects a best subset of %d of %d predictors:\n", ab$best.size,
  ↪ length(preds)))
```

```
## ABESS selects a best subset of 4 of 7 predictors:
```

```
cf <- as.numeric(coef(ab, support.size = ab$best.size))[-1]
data.frame(predictor = preds, coef = round(cf, 3))[order(-abs(cf)), ]
```

```
##      predictor      coef
## 2 ranney4_control  0.691
## 4 union_density   0.097
## 1 citi6013        0.026
## 3 evangelical_pop -0.019
```

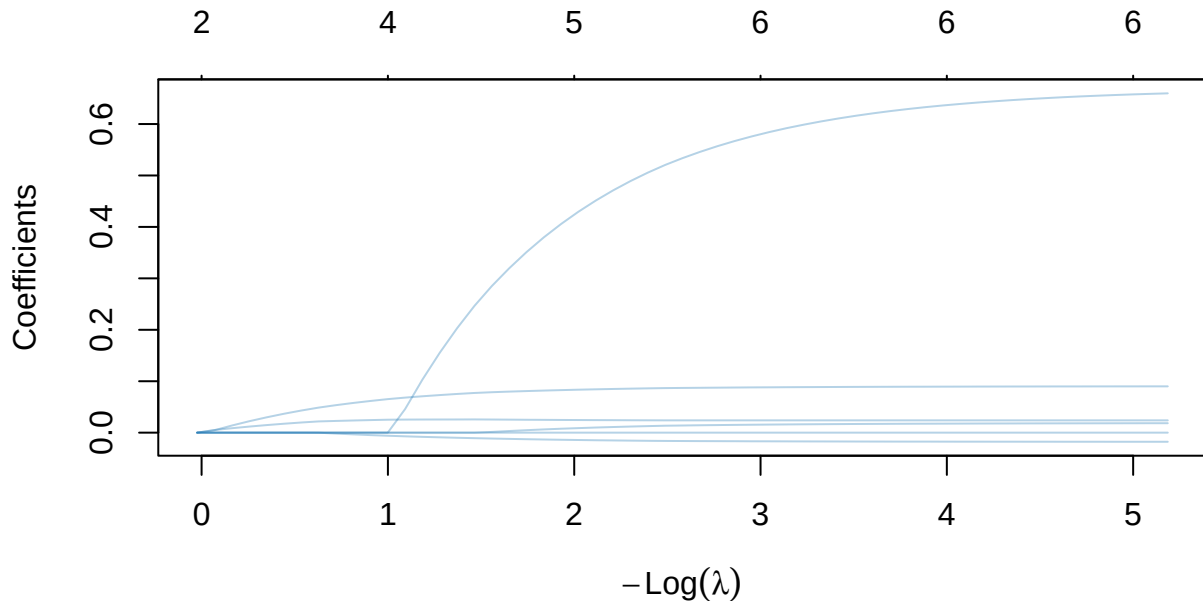


Figure 2: Lasso coefficient paths vs the penalty (log lambda); paths reaching zero are de-selected. The dashed line marks the CV-optimal penalty.

```
## 5      gini_coef  0.000
## 6      mood      0.000
## 7      gsppcap   0.000
```

Best subset and the lasso agree on the core drivers of state policy liberalism – public opinion, party control, and organized labor – reached by direct search rather than shrinkage.

7 Takeaway

This example illustrates **regularization in a small-sample design**. State-politics regressions have few observations (50 states) relative to their predictors, so ordinary regression over-fits; lasso/elastic-net (`glmnet`) and best-subset selection (`abess`) predict new states more accurately **and** name the compact set of characteristics that drive policy liberalism.

8 Recommended exercises

1. Add a couple more state characteristics from the Correlates of State Policy data and see whether the held-out gain over OLS grows as the predictor-to-state ratio rises.
2. Compare `lambda.min` and `lambda.1se` predictions: how much accuracy does the sparser 1-SE model give up?
3. Refit predicting a *specific* policy (e.g., minimum wage) instead of the liberalism index and report which predictors the lasso keeps.